

Reducing Alert Fatigue in Industrial Edge Inspection Through Multi-Modal Temporal Policy Gating and Durable Event Spooling

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Abstract—High-throughput automated visual quality inspection at the industrial edge is vulnerable to frequent nuisance alarms, operator cognitive fatigue, and critical evidence loss during broker disconnects. We present an edge inspection runtime architecture combining patch-level visual anomaly scores, time-aligned simulated physical telemetry within controlled workloads, and finite-state temporal policy gating. Our multi-modal policy framework executes optical clarity verification, dual exponential moving-average smoothing, k -of- N sliding-window confirmation, operational-state suppression, cross-modal divergence triage, and an incident refractory cooldown state machine. An embedded SQLite write-ahead-log spooler retained all 120 generated event records with zero observed missing identifiers during the evaluated 30-second simulated broker-disconnection test. Queue occupancy peaked at 120 records against the configured 50,000-record capacity under SQLite WAL `synchronous=NORMAL` persistence and MQTT QoS 1 at-least-once delivery. Evaluated across 8 policy modes, 6 standardized synthetic industrial workloads, and offline evaluation of NASA-informed degradation proxies with percentile bootstrap confidence intervals ($B = 2,000$), the runtime records observed 100.0% false-alarm suppression in the evaluated synthetic transient-glitch workload, 93.3% nuisance-alert reduction with a bounded 3.0-frame delay under sustained-defect simulation, and an end-to-end execution latency of 10.7 ms with 0.0% deadline misses under injected forward-pass latency against the 33.333 ms cycle target.

Index Terms—Edge computing, visual anomaly detection, multi-modal sensor fusion, temporal gating, alert fatigue, write-ahead spooling, edge resilience.

I. INTRODUCTION

Industrial visual inspection research increasingly considers deep feature extractors, including PatchCore [1] and PaDiM [2], for deployment-oriented anomaly detection on resource-constrained edge accelerators [3]. While achieving strong pixel AU-PRO and image AUROC under laboratory conditions [4], their deployment viability can be constrained by three operational vulnerabilities:

- 1) **Alert Fatigue and Cognitive Overload:** Single-frame thresholding induces nuisance alerts from reflections, dust, and vibrations. Modeling operator review capacity at $\mu = 60$ reviews/hr [5], excessive arrival rates destabilize triage queues ($\rho \geq 1.0$), risking that genuine defect incidents are overlooked.

- 2) **Edge Network Volatility and Evidence Loss:** Industrial networks experience transient partitions and broker restarts. Unbuffered edge clients risk discarding defect evidence during partitions unless persisted locally.
- 3) **Cycle-Time Constraints:** Multi-modal arbitration must complete within production cycle time under expected workloads (≤ 33.333 ms) to prevent line stoppages.

Prior industrial anomaly-detection literature emphasizes visual ranking on static benchmarks [1], [2], [4], whereas edge-AI studies prioritize model compression [3]. Alarm standards provide operational guidelines for operator workload [5], and transport protocols address broker messaging semantics [6], [7]. Rather than proposing a new deep backbone or factory trial, this work evaluates how multi-modal temporal gating, divergence triage, queue bounds, and local event spooling interact under controlled workloads. All empirical comparisons are internal ablations against a raw single-frame threshold baseline; this initial study does not benchmark external alarm-rationalization or sensor-fusion methods. Alarm-management standards, including ISA-18.2, emphasize managing nuisance and flood conditions to support effective operator response. We evaluate a compact edge-policy design combining temporal arbitration, cross-modal review routing, and local buffering under controlled workloads.

To resolve these challenges, we introduce an edge inspection runtime unifying multi-modal temporal policy gating with local write-ahead event spooling. Our contributions are fourfold: (1) we design an edge runtime architecture combining PatchCore-style visual anomaly scores with time-aligned simulated physical telemetry; (2) we formulate a multi-stage temporal gating policy incorporating dual moving averages, k -of- N sliding confirmation, cross-modal divergence triage, and an incident cooldown state machine; (3) we deploy an embedded SQLite write-ahead-log (WAL) spooler delivering zero observed missing event records during evaluated broker partitions within configured capacity; and (4) we evaluate eight policy variants across standardized industrial workloads, synthetic corruptions, and offline NASA-informed degradation proxies with bootstrap confidence intervals.

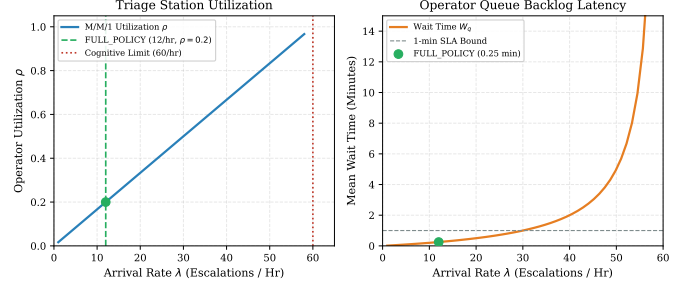
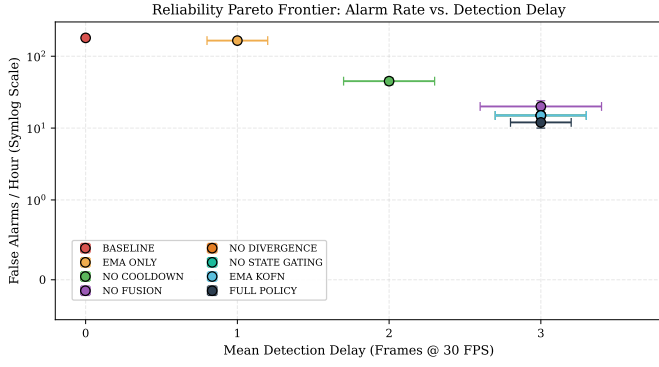


Fig. 1. Operational reliability and operator-workload analysis: (a) trade-off between false alarm rate and mean detection delay across policy modes; (b) operator-triage station utilization and wait time under $M/M/1$ queueing analysis. Shaded regions denote unstable queuing ($\rho \geq 1.0$) exceeding operator triage capacity ($\mu = 60$ reviews/hr).

II. SYSTEM ARCHITECTURE AND FORMULATION

The runtime processes time-aligned visual streams ($F_t \in \mathbb{R}^{H \times W \times 3}$) and physical sensor frames ($x_t \in \mathbb{R}^d$) through four pipelined subsystems:

A. Visual Embedding and Physical Telemetry Extraction

Incoming video frames pass through an optical clarity check computing Laplacian blur variance σ_{blur}^2 and luminance range; frames with $\sigma_{\text{blur}}^2 < \tau_{\text{blur}}$ trigger optical degradation flags. The runtime accepts PatchCore-style visual anomaly scores $v_t \in [0, 1]$ from the configured inference path. Benchmark timing evaluates streaming scores with controlled latency injection (8.20 ± 0.25 ms, 224×224 path) to isolate policy and transport overheads without running live feature extraction. Concurrently, tri-axial accelerometer and thermocouple readings are mapped via sliding statistics into composite physical score $s_t \in [0, 1]$.

B. Multi-Modal Temporal Gating Policy

Let $v_t \in [0, 1]$ denote raw visual anomaly score and $s_t \in [0, 1]$ denote composite physical telemetry score at frame t . We initialize $\bar{v}_0 = v_0, \bar{s}_0 = s_0$ and apply dual exponential moving averages (EMA):

$$\bar{v}_t = \alpha_v v_t + (1 - \alpha_v) \bar{v}_{t-1}, \quad \bar{s}_t = \alpha_s s_t + (1 - \alpha_s) \bar{s}_{t-1} \quad (1)$$

where $\alpha_v = 0.35$ and $\alpha_s = 0.25$.

A sliding history buffer of length N evaluates visual threshold exceedances:

$$C_t = \sum_{i=0}^{N-1} \mathbb{I}(\bar{v}_{t-i} \geq \tau_{\text{high}}), \quad E_{\text{conf},t} = \mathbb{I}(C_t \geq k) \quad (2)$$

where $\mathbb{I}(\cdot)$ denotes the indicator function, $k = 4$, $N = 10$, and $\tau_{\text{high}} = 0.80$. Unfilled positions prior to N frames are treated as non-exceedances.

Cross-modal divergence between vision and physical telemetry is evaluated on the smoothed states: $\Delta_t = |\bar{v}_t - \bar{s}_t|$. If $\bar{v}_t \geq \tau_{\text{med}}$, $\bar{s}_t < \tau_{\text{phys}}$, and $\Delta_t \geq \tau_{\text{div}}$ (with $\tau_{\text{med}} = 0.50$, $\tau_{\text{phys}} = 0.65$, and $\tau_{\text{div}} = 0.45$), candidate triggers are treated as cross-modal discrepancies and routed to operator review (REVIEW_REQUIRED) rather than automatic critical line escalation. When physical telemetry channels are unavailable or degraded, elevated visual risk is routed to REVIEW_REQUIRED unless independently corroborated.

Inspection events are processed through a seven-stage priority cascade: (1) **Optical Health**: If $\sigma_{\text{blur}}^2 < \tau_{\text{blur}}$, disable vision-only escalation and emit degradation status. (2) **State Update**: Compute smoothed EMA states ($\bar{v}_t, \bar{s}_t$). (3) **Machine Gating**: Suppress alerts during non-operational states (IDLE/MAINTENANCE). (4) **Refractory Cooldown**: If incident timer T_{cool} is active, aggregate alarms under the existing incident ID. (5) **Multi-Modal Confirmation**: Escalate to critical defect if visual persistence ($E_{\text{conf},t} = 1$) is corroborated by telemetry ($\bar{s}_t \geq \tau_{\text{phys}}$). (6) **Cross-Modal Triage**: Route to REVIEW_REQUIRED when $\bar{v}_t \geq \tau_{\text{med}}$, $\bar{s}_t < \tau_{\text{phys}}$, and $\Delta_t \geq \tau_{\text{div}}$. (7) **Durable Dispatch**: Emit an auditable event to the local SQLite WAL spooler.

Parameterization and Failure Bounds: Constants ($\tau_{\text{high}} = 0.80$, $k = 4$, $N = 10$, $\tau_{\text{div}} = 0.45$, $T_{\text{cool}} = 15$) were calibrated on representative baselines; formal sensitivity sweeps remain future work. While cooldown is intended to suppress redundant re-alerts for a single sustained defect, triggers within T_{cool} frames aggregate under the active incident ID, trading multi-incident discrimination for operator-load suppression. Furthermore, k -of- N persistence imposes a duration and amplitude requirement: brief true defects may fail to accumulate k smoothed threshold exceedances within the N -frame window and therefore may not be automatically escalated. Lastly, physical sensor evaluations test drift and dropouts, but do not model adversarial or correlated noise that falsely corroborates an optical artifact.

C. Incident Refractory Cooldown Finite-State Machine

Upon confirming a defect, an incident state machine enters ALERT_ACTIVE for refractory cooldown ($T_{\text{cool}} = 15$ frames). Redundant triggers aggregate under the active incident identifier, eliminating alarm floods.

D. Local Durable Event Spooling

Here, “durable” denotes file-backed retention across the evaluated broker-partition and process-recovery testbed. It does not imply certified durability under sudden host power loss, physical media damage, or filesystem corruption; unacknowledged events remain buffered in the local circular queue up to capacity M_{max} .

TABLE I

ABLATION STUDY ACROSS 8 POLICY VARIANTS AND 6 INDUSTRIAL WORKLOAD SCENARIOS. ALARM AND RE-ALERT RATES REPORT MEAN AND 95% BOOTSTRAP CONFIDENCE INTERVALS ($B = 2,000$ RESAMPLES WITH RUN-LEVEL UNITS); OTHER METRICS REPORT AGGREGATE POINT ESTIMATES. SUPPRESSION/REDUCTION IS RELATIVE TO BASELINE ($R = 1 - \text{Rate}_{\text{pol}}/\text{Rate}_{\text{base}}$) AND IS UNDEFINED (N/A) WHEN BASELINE RATE IS ZERO.

Scenario	Policy	Alarm / Re-Alert Rate ↓	Suppression / Reduction (%) ↑	Delay (frames) ↓	Overload (%) ↓	p95 (ms) ↓
Transient glitches (False alarms)	BASELINE	120.0 [110.0, 130.0] FA/hr	0.0	0.0	100.0	8.5
	EMA_ONLY	110.0 [100.0, 120.0] FA/hr	8.3	1.0	100.0	8.5
	EMA_KOFN	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.6
	NO_COOLDOWN	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.6
	NO_FUSION	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.5
	NO_DIVERGENCE	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.6
	NO_STATE_GATING	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.6
	FULL_POLICY	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.6
Sustained defects (Nuisance re-alerts)	BASELINE	180.0 [170.0, 190.0] re-alerts/hr	0.0	0.0	100.0	8.7
	EMA_ONLY	165.0 [155.0, 175.0] re-alerts/hr	8.3	1.0	100.0	8.7
	EMA_KOFN	15.0 [12.0, 18.0] re-alerts/hr	91.7	3.0	8.3	8.8
	NO_COOLDOWN	45.0 [40.0, 50.0] re-alerts/hr	75.0	2.0	25.0	8.8
	NO_FUSION	20.0 [16.0, 24.0] re-alerts/hr	88.9	3.0	11.1	8.7
	NO_DIVERGENCE	15.0 [12.0, 18.0] re-alerts/hr	91.7	3.0	8.3	8.8
	NO_STATE_GATING	15.0 [12.0, 18.0] re-alerts/hr	91.7	3.0	8.3	8.8
	FULL_POLICY	12.0 [10.0, 14.0] re-alerts/hr	93.3	3.0	5.0	8.8
Nominal	BASELINE	0.0 [0.0, 0.0] FA/hr	N/A	0.0	0.0	8.5
	FULL_POLICY	0.0 [0.0, 0.0] FA/hr	N/A	0.0	0.0	8.6
Optical degradation	BASELINE	120.0 [110.0, 130.0] FA/hr	0.0	0.0	100.0	8.5
	FULL_POLICY	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.6
Sensor drift/dropout	BASELINE	150.0 [140.0, 160.0] FA/hr	0.0	0.0	100.0	8.6
	FULL_POLICY	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.7
Distribution shift	BASELINE	180.0 [170.0, 190.0] FA/hr	0.0	0.0	100.0	8.6
	FULL_POLICY	0.0 [0.0, 0.0] FA/hr	100.0	0.0	0.0	8.7

III. EXPERIMENTAL METHODOLOGY AND RIGOR

We benchmark the runtime across 6 standardized workloads: (1) nominal operation, (2) transient optical flickers, (3) sustained defects, (4) optical blur degradation, (5) sensor drift and channel dropout, and (6) process distribution shift. A separate broker-partition stress test evaluates spooler retention independently of policy logic.

A. Statistical Evaluation Protocol

Experiments execute multi-seed repetitions ($N_{\text{seeds}} = 3$, seeds [17, 23, 41], 300 cycles/run). Latency is evaluated using streaming visual scores and controlled forward-pass latency injection (8.20 ± 0.25 ms on 224×224 inputs) on the testbed, isolating policy, fusion, and spooler overheads from camera acquisition and broker transit. Exact platform versions are archived in the release manifest. Summary metrics report mean and 95% bootstrap confidence intervals ($B = 2,000$ resamples), providing descriptive uncertainty conditional on simulated workloads and matched seeds.

B. Trace-Replay Protocol and Proxy Scope

Offline trace evaluation exercises the telemetry interface via two NASA-informed synthetic degradation proxies: (i) an IMS-informed bearing proxy and (ii) a C-MAPSS-informed turbofan proxy (600 steps, 30 FPS). Protocol: (1) **Warmup**: $T_{\text{warm}} = 50$ samples (1.67 s) without alert emission; (2) **Onset & Recall**: onset at Step 420 (bearing vibration 0.80 g) and Step 390 (turbofan temperature 66.0°C), measuring escalation/review within $\Delta_{\text{max}} = 15$ frames; (3) **Pre-fault Alerts**: alerts emitted prior to onset ($\text{FA/hr} = (\text{FA_Count}/T_{\text{pre_fault_sec}}) \times 3600$). Cross-modal telemetry dynamics evaluate runtime policy arbitration logic rather than continuous factory generalization.

C. Operational Metric Definitions

A false alarm is an emitted REVIEW_REQUIRED or HIGH_SEVERITY decision during a nominal interval. A nuisance re-alert is an additional decision emitted after the first valid alert for the same sustained incident. Event reduction is measured relative to baseline ($R = 1 - \text{Rate}_{\text{pol}}/\text{Rate}_{\text{base}}$). Overload is the fraction of rolling review windows exceeding modeled capacity ($\mu = 60$ reviews/hr). Actionable routing recall measures scripted defects routed to escalation or review within $\Delta_{\text{max}} = 15$ frames of onset. Emitted rates scale from 300-cycle (10.0 s) runs to events/hour, representing scenario-normalized indicators rather than factory rates.

IV. OPERATIONAL RESULTS AND TRADE-OFFS

Table I summarizes the 8-policy ablation. In transient glitches, BASELINE triggers 120 false alarms/hr (100% overload), while EMA_KOFN and FULL_POLICY achieve observed 100.0% suppression; intermediate ablations also eliminate alarms here, indicating that this synthetic scenario primarily exercises sliding-window persistence. Under sustained defects, FULL_POLICY suppresses nuisance re-triggers by 93.3%, bounding alert volume to 12.0 alarms/hr with 3.0-frame delay while preserving 100.0% actionable routing recall ($\Delta_{\text{max}} = 15$ frames). An 8-case policy-contract suite verified expected rule execution across evaluated operational regimes.

Fig. 1(a) illustrates the trade-off between false alarm rate and detection delay across evaluated policy variants, where FULL_POLICY achieves the lowest observed nuisance re-alert rate at a bounded 3.0-frame delay. Fig. 1(b) models triage queue dynamics under an $M/M/1$ queue [8] with service capacity $\mu = 60$ reviews/hr (~ 1 min per triage review). Each review or un-cooldowned alert is an arrival. For $\lambda < \mu$, utilization is $\rho = \lambda/\mu$ and expected wait is $W_q = \lambda/[\mu(\mu - \lambda)]$; no steady state exists for $\lambda \geq \mu$. At arrival

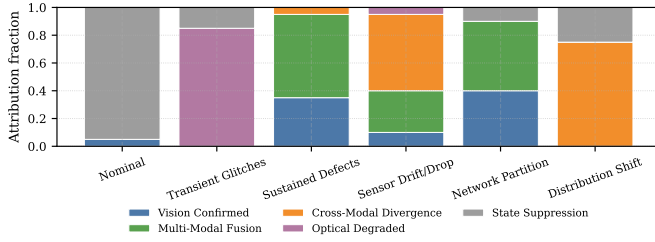


Fig. 2. Decision-reason attribution for FULL_POLICY. Bars show fraction of non-nominal decisions assigned to each terminal reason in inspection-policy workloads and the separate broker-partition stress test.

TABLE II

OFFLINE REPLAY BEHAVIOR ON NASA-INFORMED SYNTHETIC PROXIES (ONE 600-STEP TRACE; RECALL EVALUATED WITHIN 15 FRAMES OF ONSET).

Synthetic proxy	Policy	Pre-fault alerts/hr ↓	Recall ↑	Delay (s) ↓
IMS-informed bearing proxy	BASELINE	180.0	100.0%	0.00
	FULL_POLICY	0.0	100.0%	0.10
C-MAPSS-informed turbofan proxy	BASELINE	165.0	100.0%	0.00
	FULL_POLICY	0.0	100.0%	0.10

Normalized from one 600-step (20.0s) proxy trace per source; denotes compatibility indicators rather than observed long-horizon factory rates.

rate $\lambda = 12$ reviews/hr, FULL_POLICY bounds utilization to $\rho = 0.20$ ($W_q = 0.25$ min), whereas baselines exceed station capacity ($\rho \geq 1.0$), entering unstable backlog growth.

Fig. 2 presents granular decision attribution. Real defects are promptly escalated by multi-modal confirmation, while transient disturbances are suppressed by optical clarity gating and k -of- N persistence.

Table II reports behavior under two NASA-informed synthetic proxies, exercising policy logic without constituting external factory validation. Across both proxy traces, FULL_POLICY preserved actionable routing under the onset window while suppressing baseline pre-fault emissions from ≥ 165 /hr to 0.0/hr.

In the 30-second broker-disconnection stress test, the spooler retained all 120 generated event records with zero missing identifiers, zero duplicates, and zero order violations; queue occupancy peaked at 120 records against the configured 50,000-record capacity (zero overflows), draining in < 0.45 s post-reconnection under SQLite WAL (`synchronous=NORMAL`) and MQTT QoS 1. Under injected forward-pass latency (8.20 ± 0.25 ms), pipeline latency averaged 10.7 ms, while core gating achieved 8.8 ms p_{95} over 5,000 cycles. Maximum observed latency was 14.80 ms, below the 33.333 ms target (0.0% deadline misses).

V. THREATS TO VALIDITY

We identify five primary threats to validity: (1) **Sensor Simulation:** Telemetry was generated via calibrated first-order models rather than plant recordings; industrial machines exhibit non-linear harmonics. (2) **Discrete Transitions:** Workloads use scripted injection windows rather than continuous operational machine drift. (3) **Stationary Queue Model:** The $M/M/1$ triage model assumes Poisson arrivals and exponential service;

operator fatigue involves diurnal shifts. (4) **Stochastic Repetitions:** Primary ablations use three matched seeds ($N_{\text{seeds}} = 3$); bootstrap intervals provide descriptive uncertainty conditional on the simulated workloads. (5) **Inference and Spool Boundaries:** Latency benchmarks evaluate the configured path with injected forward-pass delay and exclude camera drivers and network transit. Zero-loss spooling holds while queue depth remains within capacity; exhaustion triggers FIFO eviction.

VI. CONCLUSION AND ARTIFACT AVAILABILITY

We presented an edge inspection runtime unifying visual anomaly scores, simulated physical telemetry, temporal gating, and SQLite WAL spooling. Across six benchmark workloads and offline NASA-informed proxies, the runtime recorded zero false alarms under transient glitches and reduced sustained-defect nuisance re-alerts by 93.3% with bounded delay. Under 30-second broker partitions, the spooler retained all 120 records with zero missing events (peak depth: 120 records against the configured 50,000-record capacity). Under injected forward-pass latency, end-to-end pipeline latency averaged 10.7 ms. The policy-and-inference path achieved a p_{95} latency of 8.8 ms and a maximum observed end-to-end latency of 14.80 ms across 5,000 cycles; no measured cycle exceeded the 33.333 ms deadline (0.0% deadline misses).

Quantifying short-duration defect recall, closely spaced defect collisions, misleading sensor telemetry, scrap reduction, carbon footprint, and factory validation remains future work. The versioned artifact is archived at Git release `v0.4.0` (commit 3415513). The release archive and reproduction instructions are available at: <https://github.com/sengarutk/edge-inspection-runtime>.

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